Antonio Alvarez Valdivia

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EDUCATION

Purdue University – West Lafayette, Indiana — Graduation Date: Dec 2025
- Ph.D. in Mechanical Engineering — *Advisor: Laura H. Blumenschein*
  Dissertation: Design, Control and Usability Analysis of Ubiquitous Haptic Interfaces for Human-Machine Interaction

Iowa State University – Ames, Iowa — Graduation Date: May 2021
- Bachelor of Science in Mechanical Engineering

Des Moines Area Community College – Ankeny, Iowa — Graduation Date: May 2018
- Pre-Engineering (Transfer Program)

EXPERIENCE

Postdoctoral Research Associate — Aug 2026 – Present
Northeastern University
Advisor: Kris Dorsey

Postdoctoral Research Associate — Jan 2026 – Aug 2026
Massachusetts Institute of Technology, Lincoln Laboratory
Advisor: Nathaniel Hanson

Research Assistant — Aug 2021 – Dec 2025
Purdue University, Mechanical Engineering
Advisor: Laura H. Blumenschein

Research Intern — May 2025 – Aug 2025
Massachusetts Institute of Technology, Lincoln Laboratory

Visiting Researcher — Oct 2023 – Nov 2023
Virginia Tech, Mechanical Engineering
Advisor: Dylan Losey

Undergraduate Research Assistant — Jun 2018 – Dec 2020
Iowa State University, Mechanical Engineering
Advisor: Jaime J. Juarez

Summer Undergraduate Research Assistant — May 2019 – Aug 2019
University of Pennsylvania, Mechanical Engineering and Applied Mechanics
Advisor: Kevin T. Turner

RESEARCH INTERESTS

My research focuses on human-centered robotic systems that integrate sensing, actuation, and adaptable mechanical design to interact effectively with people and complex environments. My interests span haptics, soft and wearable robotics, human–robot interaction, and embodied intelligence, with applications ranging from physical communication and assistance to field-deployable systems for search and rescue.

PUBLICATIONS

Journal Articles:

J11 **Alvarez Valdivia, A.,** McFarland, C., Reeve, R., Dhawan, A., Council, C., Richardson, M., McGuinness, M., Hanson, N. (2026) *SPROUT: The Open-Source Soft Growing Robot for Search and Rescue*. (Under Review).

J10 Aubin, C., Yuen, M., **Alvarez Valdivia, A.**, Dorsey, K.L. (2026) *The sensor does (not) stand alone: sensor performance and accessibility for soft robotic systems.* (Under Review).

J9 Saldarriaga, B., Kim, S., Wang, S., Montrone, F., **Alvarez Valdivia, A.**, Scholnik, L., Goel, S., Mangan, A., Bergeles, C., Morimoto, T.K., Blumenschein, L.H. (2026) *Cross-Platform Reproducibility of Laser-Welded Thin TPU Films for Soft Robotic Manufacturing.* (Under Review).

J8 **Alvarez Valdivia, A.**, Reeve, R., Dhawan, A., McFarland, C., Council, C., McGuinness, M., Hanson, N. (2026) *Gotta Grow Fast: Design and Benchmarking of a Tip Mount for High-Speed Vine Robots*. IEEE Robotics and Automation Letters. DOI: 10.1109/LRA.2026.3709052.

J7 **Alvarez Valdivia, A.**, Christie, B.H., Losey, D.P., and Blumenschein, L.H. (2026) *A Modular Haptic Display with Reconfigurable Signals for Personalized Information Transfer.* IEEE Transactions on Haptics. DOI: 10.1109/TOH.2026.3650759.

J6 McFarland, C.*, **Alvarez Valdivia, A.***, Taher, S., Hanson, N., McGuinness, M. (2025) *On Steerability Factors for Growing Vine Robots. arXiv preprint arXiv:2510.22504*.

J5 Huang, B., Wang, Z., Cheng, Q., Ren, S., Cai, H., **Alvarez Valdivia, A.**, Mahadevan, K., Wigdor, D. (2025) *AeroHaptix: A Wearable Vibrotactile Feedback System for Enhancing Collision Avoidance in UAV Teleoperation*. IEEE Robotics and Automation Letters. DOI: 10.1109/LRA.2025.3548866.

J4 Habibian, S.*, **Alvarez Valdivia, A.***, Shailly, Blumenschein, L.H. and Losey, D.P. (2024) *A Survey of Communicating Robot Learning during Human-Robot Interaction*. International Journal of Robotics Research. DOI:10.1177/02783649241281369.

J3 **Alvarez Valdivia, A.**, Habibian, S., Mendenhall, C.A., Fuentes, F., Shailly, R., Losey, D.P. and Blumenschein, L.H. (2023) *Wrapping Haptic Displays Around Robot Arms to Communicate Learning*. IEEE Transactions on Haptics. DOI: 10.1109/TOH.2023.3240400 . ***(2024 IEEE Transactions on Haptics Best Application Paper Award).***

J2 Shabaniverki, S., **Alvarez Valdivia, A.** and Juárez, J.J. (2021) *3D printed self-propelled composite floaters*. Smart Materials and Structures. DOI: 10.1088/1361-665X/ac01a9.

J1 Shabaniverki, S., **Alvarez Valdivia, A.** and Juárez, J.J. (2019) *Portable imaging viscometry for quantitative complex fluid measurements*. Experimental Thermal and Fluid Science. DOI: 10.1016/j.expthermflusci.2019.05.009.

Refereed Conference Articles:

C7 **Alvarez Valdivia, A.**, Kestrica, A., Doran, E., Cantley, L., Parker, N., Evans, B., Hanson, N. (2026) *Deploying Large-Scale Underwater Sensor Arrays with Growing Vine Robots.* (Under Review).

C6 Godoy, S. L., McFarland, C., Val, M., **Alvarez Valdivia, A.**, Hanson, N., & McGuinness, M. (2026). Distributed Acoustic Localization Array Deployed Using a Soft Everting Vine Robot. 2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS).

C5 Laudenslager, A.E., **Alvarez Valdivia, A.,** Hanson, N., and McGuinness, M. (2026) Evaluating Accuracy of Vine Robot Shape Sensing with Distributed Inertial Measurement Units. 2026 11th IEEE RAS/EMBS International Conference on Biomedical Robotics and Biomechatronics (BioRob).

C4 Wang, S., Frias-Miranda, E., **Alvarez Valdivia, A.**, and Blumenschein, L.H. (2025) *Anisotropic Stiffness and Programmable Actuation for Soft Robots Enabled by an Inflated Rotational Joint*. 2025 IEEE International Conference on Robotics and Automation (ICRA). DOI: 10.1109/ICRA55743.2025.11127236.

C3 **Alvarez Valdivia, A.**, Rezqalla, Mohammad A., Swann, Sarah E., and Blumenschein, L.H. (2023) *Soft Growing Pin for High-Extension Shape-Changing Displays*. 2024 IEEE International Conference on Soft Robotics (RoboSoft). DOI: 10.1109/RoboSoft60065.2024.10522001.

C2 **Alvarez Valdivia, A.** and Blumenschein, L.H. (2023) *Perception of and Response to a Haptic Device as a Function of Signal Complexity*. 2023 IEEE World Haptics Conference (WHC). DOI: 10.1109/WHC56415.2023.10224490.

C1 **Alvarez Valdivia, A.**, Shailly, R., Seth, N., Fuentes, F., Losey, D.P. and Blumenschein, L.H. (2022) *Wrapped haptic display for communicating physical robot learning*. 2022 IEEE 5th International Conference on Soft Robotics (RoboSoft). DOI: 10.1109/RoboSoft54090.2022.9762210.

HONORS AND AWARDS

- 2024 IEEE Transactions on Haptics Best Application Paper Award
- 2024 Preparing Future Faculty (PFF) Program Scholar, Purdue Graduate School
- NSF Graduate Research Fellowship, 2022
- Purdue Frederick N. Andrews Fellowship, 2021
- Purdue Graduate Bridge Program Fellow, 2021
- Iowa State University MSA Academic Excellence Award, 2020
- Tau Beta Pi Engineering Honor Society, February 2020
- Ronald E. McNair Outstanding First Year Scholar, May 2019
- Ronald E. McNair Scholar, 2018

PRESENTATIONS

Invited Conference and Workshop Presentations

- “Reproducibility Lessons from an Open-Source Soft Growing Robot,” Soft Robot Manufacturing and Testing Hackathon, 2026 IEEE/RSJ International Conference on Intelligent Robots and System (IROS). Pittsburgh, PA.

Conference and Workshop Presentations

- “Perception of and Response to a Haptic Device as a Function of Signal Complexity,” Oral Presentation in 2023 IEEE World Haptics Conference (WHC). Delft, Netherlands.
- “Portable Imaging Viscometry for Quantitative Complex Fluid Measurements.” In 2020 14th ISU Symposium on Undergraduate Research and Creative Expression. Iowa State University. Ames, IA.
- “Portable Imaging Viscometry for Quantitative Complex Fluid Measurements.” in 2020 22nd Texas National McNair Research Conference. University of North Texas. Denton, TX.
- “Flexible Capacitive Force Sensors for use in Robotic Grippers,” in Summer 2019 REU Symposium. University of Pennsylvania. Philadelphia, PA.

Invited Academic Seminars

- “Soft Growing Robots for Confined Space Ingress, Sensing, and Rescue,” Advanced Concepts Committee (ACC) Showcase, MIT LL, 2026.
- “Soft Haptic Interfaces for Human-Machine Interaction,” Robotics @ Notre Dame Seminar, University of Notre Dame, 2025.

General Audience Presentations

- “Designing Soft Robots for Human and Environmental Interactions,” Invited Podcast Presentation on Talking Robotics. October 2026.

Poster Presentations

1. “Soft Growing Pin for High-Extension Shape-Changing Displays,” in 2024 IEEE International Conference on Soft Robotics (RoboSoft). San Diego, CA.
2. “Wrapping Haptic Displays Around Robot Arms to Communicate Learning,” in 2023 IEEE International Conference in Robotics and Automation (ICRA). London, UK.
3. “Wrapping Haptic Displays Around Robot Arms to Communicate Learning,” in 2023 Inaugural ICON Student Research Conference. Purdue University. West Lafayette, IN.
4. “Wrapped haptic display for communicating physical robot learning,” in 2022 IEEE 5th International Conference on Soft Robotics (RoboSoft). Edinburgh, Scotland.
5. “555-timer Flexible Circuit,” in 2019 McNair Program Research Symposium. Iowa State University. Ames, IA.
6. “Portable Imaging Viscometry for Quantitative Complex Fluid Measurements,” in Summer 2018 REU Symposium. Iowa State University. Ames, IA.

RESEARCH FUNDING

ICE-EARS: Ice Cap Evertible Acoustic Reconnaissance System **$125,000**

Antonio Alvarez Valdivia (PI, MIT LL), Nathaniel Hanson (Co-PI, MIT LL), Ben Evans (Co-PI, MIT LL). Office of the Under Secretary of War for Research and Engineering (OUSW R&E); Apr 2026 - Sept 2026. Awarded March 2026.

PATENTS

- Distributed Fiber-Based Sensor Arrays for Growing Vine Robots. U.S. Provisional Patent, U.S. Provisional Patent, 64/111,429, filed Jul 14, 2026. Patent pending.
- Triangular Sensor Mount for Robots that Grow via Eversion. U.S. Provisional Patent, U.S. Provisional Patent, 63/971,879, filed Jan 30, 2026. Patent pending.

TEACHING

Undergraduate Teaching Assistant - Iowa State University 2020

- Grading duties for assignments, exams, and projects for ME 325: Mechanical Component Design.
- Held office hours for 30-40 students

Summer School Teacher - Ames Community School District 2017

- Created *Engineering 101* course curriculum.
- Taught 6-week summer school course for 1st - 6th grade students.

MENTORING

Graduate Students Mentored

Andrew Mejia Hernandez	Northeastern University, 2026
Lily Scholnik	Purdue University 2023-2025
Carly Mendenhall (CBRE GWS)	Purdue University 2022-2024

Undergraduate Students Mentored

Olivetan Millien	Northeastern University, 2026
Jaclyn Cohen	Northeastern University, 2026
Ravindu Karunaratne	Northeastern University, 2026
Antonio Guizar Jaurena	Iowa State University, 2026
Mohammad Rezqalla	Purdue University, 2023-2025
Ryan Carlson	Purdue University, 2025
Carissa Lukac	Purdue University, 2023-2024
Sarah Swann (Seagate Technologies)	Purdue University, 2024
Aadit Kumar	Purdue University, 2024
Ria Singhal (Cowboy Space)	Purdue University, 2023
Norawish Lohitnavy (ASML)	Purdue University, 2022
Andy Ding (Anduril Industries)	Purdue University, 2022

PROFESSIONAL SERVICE

Peer Reviewer

- IEEE Robotics and Automation Letters
- ACM Transactions on Human-Robot Interaction
- IEEE Control Systems Magazine
- 2027 IEEE International Conference on Robotics and Automation (ICRA)
- 2026 IEEE International Conference on Robotics and Automation (ICRA)
- 2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

- 2025 IEEE International Conference on Soft Robotics (RoboSoft)
- 2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

Organization Committee | HADR & IROS 2026

- Helped organize the Robotics and Autonomous Systems in Humanitarian Assistance and Disaster Relief (HADR) Workshop at 2026 IROS

General Co-Chair | 3rd ICON Student Research Conference Jun 2024 – Feb 2025

- Student-run conference at Purdue's Institute for Control, Optimization and Networks (ICON).
- Led conference organization, program, and logistics.
- Secured $4,000 in sponsorship from the IEEE CSS Diversity, Outreach & Development Activities Board and $4,000 from Purdue's ICON to fund the conference.

Logistics Co-chair | 2nd ICON Student Research Conference Oct 2023 – April 2024

- Student-run conference at Purdue's Institute for Control, Optimization and Networks (ICON).
- Assisted with location, programming, and banquet reservations.

LEADERSHIP

Vice President | OMEGA (Mechanical Engineering Graduate Association) May 2024 – May 2025

- Represented OMEGA at Purdue Engineering Graduate Student Association Committee Meetings.
- Organized major social and networking events for ME graduate students, faculty, and staff.

Diversity Officer | OMEGA (Mechanical Engineering Graduate Association) Jan 2022 – May 2023

- Organized major social and networking events for ME graduate students, faculty, and staff.
- Represented Purdue ME department in graduate school diversity recruitment programs.

OUTREACH

Volunteer | Purdue Women in Engineering Program Nov 2021 – Dec 2024

- Assisted in engineering workshops and short courses directed by Prof. Blumenschein (RAAD Lab) in programs hosted by Purdue's Women in Engineering (WiE).

International Student Orientation Leader | Iowa State University Jun 2019 - Aug 2019

- Led orientation for groups of international students for the International Students and Scholars Office.
- Assisted with registration, English placement, and immigration processes.
- Exemplified respectful practices and communication with multicultural, international students.
- Organized group activities and discussions facilitation

Peer Mentor | Iowa State University Sep 2018 – May 2021

- Engineering Admissions Partnership Program for Engineering Student Services.
- Communicate with prospective transfer students.
- Network with industry partners and student organizations to feature articles in monthly newsletters.

OPEN-SOURCE MATERIALS

SPROUT: Soft Pathfinding Robotic Observation Unit
Open-source robotic platform with design files, fabrication instructions, and technical documentation.
https://sprout-mitll.github.io/sprout/

Gotta Grow Fast: Design and Benchmarking of a Tip Mount for High-Speed Vine Robots
Open-source CAD files and documentation for a tip-mounted camera assembly.
https://sprout-mitll.github.io/tip_mounts/CAD/tip_mount/

A Survey of Communicating Robot Learning During Human-Robot Interaction
Open-source code for software used in case study, including robot learning algorithm, Franka Research 3 control, haptic and augmented reality interfaces (Arduino, Unity/C#, HoloLens), and experimental data processing.
https://github.com/VT-Collab/communicating-robot-learning

SELECTED RESEARCH COVERAGE

- University of Notre Dame, "Fighting for Robotic Search and Rescue," Notre Dame Stories, Sep. 2026. [Multimedia Feature].
- NBC, University of Notre Dame, "How Robots Can Help in Search and Rescue," YouTube, Sep. 11, 2026. [Video Feature].
- MIT News, "A Flexible Robot Can Help Emergency Responders Search Through Rubble," Apr. 2, 2025. [News Feature].
- Virginia Tech News, "Robots Collaborating with Humans," Nov. 2023. [Video Feature].